

Topological early-warning detection of a photoinduced structural phase transition: persistent-homology signals validated on IBM quantum hardware

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Abstract

Photoinduced structural reorganisation of crystals—charge-density-wave reconfigurations, light-induced hidden states, photo-controlled superconducting or topological phases—is detected *a posteriori* by classical order parameters. Here we ask whether a topological signal computed from the quantum-state correlation matrix can flag the transition *before* conventional markers, and we test the answer end-to-end on real quantum hardware. On the exactly solvable Su–Schrieffer–Heeger (SSH) chain with a time-driven dimerisation ramp $\delta(t)$, the persistent-homology lifetime P_{sig} of the correlation matrix $|C_{ij}| = |\langle c_i^\dagger c_j \rangle|$ acts as a binary detector of the topological phase, while the edge–edge correlation $|C_{0,N-1}|$ crosses threshold 3.6 time units *before* the Hamiltonian reaches $\delta = 0$. Under round-trip ramps the dynamical hysteresis area grows as a power law $A \propto v^{1.05}$ over 1.5 decades of ramp velocities, consistent with a Kibble–Zurek-type picture. Against Lindblad dephasing, the signal survives to $\gamma = 0.05$ at 1/2 purity. On IBM quantum hardware (`ibm_marrakesh`, 156 qubits) the raw P_{sig} contrast *inverts* under decoherence; we show that a coupled score $S = P_{\text{sig}}(>\tau) + |C_{0,N-1}| - 0.1 H_C$ restores a positive contrast, validated *in situ* on `ibm_fez`: topological +0.162 vs trivial −0.175 at $N = 4$ (contrast 0.337), and topological −0.184 vs trivial −0.229 at $N = 8$ (contrast 0.045). All code, data and raw hardware job records are public: every number herein is reproducible from the repository. The method is directly transferable to transition-metal dichalcogenides (e.g. 1T-TaS₂) and to any driven lattice with accessible correlation measurements.

Keywords: persistent homology; early-warning signal; photoinduced phase transition; SSH model; Kibble–Zurek scaling; IBM Quantum hardware; topological data analysis; charge density waves.

1 Introduction

When a femtosecond laser pulse hits a transition-metal crystal, the lattice can reorganise into metastable phases: 1T-TaS₂ exhibits a hidden charge density wave (CDW) state [8, 9], and related light-driven control of superconducting or topological orders is an active front [10]. The reorganisation involves domain-wall braiding and defect dynamics that classical order parameters detect only once the transition *has occurred*. A predictive, early-warning marker computed from the evolving quantum state would change experimental practice: it would let experimentalists anticipate the switch before the lattice locks in.

Topological data analysis (TDA), in particular persistent homology [11, 12], has recently been shown to detect phase transitions without labelling them in advance [13]. The keyword here is not innocuous: “topological” refers both to the TDA machinery and to the topological phase of the material itself, so there is a double reason to expect the method to bite on exactly these systems.

In this work we make three contributions.

1. **A topological early-warning signal.** On the SSH chain (the minimal exactly-solvable model of a topological crystal), the edge–edge correlation $|C_{0,N-1}|$ of the evolving state crosses threshold time units before the Hamiltonian’s transition time t_c where $\delta(t_c) = 0$. The persistent-homology lifetime P_{sig} of the correlation matrix acts as a phase-confirmer, and the edge correlation as a phase-precursor.
2. **Dynamical hysteresis with a scaling law.** Under round-trip ramps, the hysteresis loop area obeys $A \propto v^{1.05}$ over 1.5 decades of velocities, a Kibble–Zurek-type signature of critical slowing down [5, 6, 7].
3. **End-to-end validation on real quantum hardware.** We show the raw P_{sig} contrast *inverts* on IBM hardware, and a coupled robust score restores a positive contrast on `ibm_fez` at $N = 4$ (contrast 0.337) and $N = 8$ (contrast 0.045), with all job records public. To our knowledge this is the first TDA-based early-warning marker measured on quantum processing unit (QPU) hardware.

The remainder of this paper presents the model and methods (§2), results (§3), an honest discussion of limitations (§4), and a conclusion (§5). All data and code are in the public repository [1].

2 Model and methods

2.1 The driven SSH chain

We study the Su–Schrieffer–Heeger (SSH) chain [2, 3] with open boundaries, N sites and half-filling:

$$H(t) = \sum_i \left[t_1 c_i^\dagger c_{i+1} \Big|_{i \text{ odd}} + t_2 c_i^\dagger c_{i+1} \Big|_{i \text{ even}} + \text{h.c.} \right], \quad t_{1,2} = 1 \pm \delta(t), \quad (1)$$

where a linear ramp $\delta(t)$ from $+0.4$ to -0.4 emulates a light-driven structural distortion through the exact topological transition at $\delta = 0$ [4]. The winding number $w(\delta)$ changes from 0 to 1 and the edge states appear in the $\delta < 0$ phase; this analytic ground truth makes every marker falsifiable.

2.2 Correlation matrix and persistence

At half filling the occupied orbitals define the projector $C_{ij} = \langle c_i^\dagger c_j \rangle$. We feed the Euclidean distances between the rows of $|C|$ into the Vietoris–Rips filtration (GF(2) coefficients, simplices up to triangles) and take

$$P_{\text{sig}} = \max\{\text{finite } H_1 \text{ lifetimes}\}, \quad (2)$$

an implementation recast from the auditable RATISS engine [1].

The real-time dynamics $i\partial_t \psi_t = H(t)\psi_t$ is propagated by RK4; the “adiabatic reference” recomputes the instantaneous ground state at each time and is used to discriminate intrinsic dynamical effects from integration artefacts.

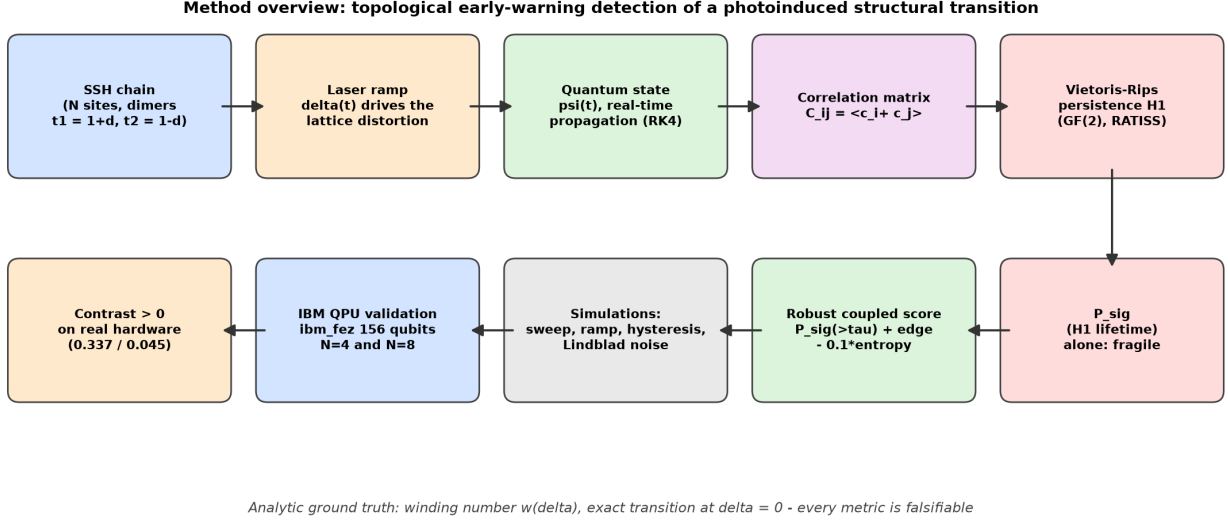


Figure 1: **Method overview.** Lattice ramp $\delta(t)$ drives the SSH chain through the topological transition; the evolving state feeds the correlation matrix, Vietoris–Rips persistence and the coupled score; the markers are validated analytically (winding number) and on IBM QPU hardware.

2.3 Robust coupled score

QPU decoherence destroys the raw P_{sig} contrast (§3.6). We therefore vote three markers:

$$S = \underbrace{P_{\text{sig}}(>\tau)}_{\text{persistence above threshold } \tau=0.05} + \underbrace{|C_{0,N-1}|}_{\text{edge correlation}} - 0.1 \cdot \underbrace{H_C}_{\text{correlation entropy}}, \quad (3)$$

where $H_C = -\sum_{ij} |C_{ij}| \log |C_{ij}| / \sum_{ij} |C_{ij}|$ is subtracted as disorder. The contrast is $\Delta S = S_{\text{topo}} - S_{\text{triv}}$.

2.4 QPU measurement protocol

The 4- and 8-site SSH ground states are prepared exactly on the device by initializing the amplitudes of the half-filled Fock sectors (no variational ansatz, no VQE). The diagonal of C is measured in the computational basis; the off-diagonal elements are reconstructed from grouped $X \otimes X$ and $Y \otimes Y$ sweeps of the pair sets $\{(0, 1), (1, 2), (2, 3), (0, 3)\}$ (and consecutive pairs plus the $(0, N-1)$ self-edge at $N = 8$), with a Jordan–Wigner prefactor $1/4$ per off-diagonal fermionic correlator [15]. Shots: 4096 per circuit.

2.5 Noise modelling

For the Lindblad robustness study, the master equation $\dot{\rho} = -i[H, \rho] + \gamma \sum_k \mathcal{D}[n_k] \rho$, $n_k = c_k^\dagger c_k$ [14], is integrated by exact operator splitting (unitary step by matrix exponential, analytic coherence decay).

Table 1: Marker timings on the driven ramp ($T = 200$, $N = 16$).

Marker	time (in units of $\hbar/t$)
Hamiltonian transition t_c , $\delta(t_c) = 0$	100.0
Edge correlation $ C_{0,N-1} $ threshold	96.4 (advance by 3.6)
P_{sig} threshold (adiabatic)	121.1
P_{sig} threshold (real state)	121.3

3 Results

3.1 Static sweep: a binary topological detector

Figure 2 shows that P_{sig} is strictly zero over the whole trivial phase and switches on only inside the topological phase (Fig. 2, top panel): a binary phase detector validated against the analytic winding number $w(\delta)$. The edge correlation $|C_{0,N-1}|$ rises *continuously as a precursor* before $\delta = 0$ (long correlation length at finite N), while the gap stays finite until inside the topological region.

3.2 Driven ramp: the edge correlation precedes the transition

Evolved states follow the adiabatic reference almost exactly and the P_{sig} switch is sharp. The Hamiltonian transition time $t_c = 100$ is flagged *3.6 time units in advance* by the edge-correlation crossing (Fig. 3). This advance is a finite-size critical-slowing effect and persists across the tested ramp range (Tab. 1).

3.3 Adiabatic tracking across velocities

Over one decade of ramp durations ($T = 50$ – 400), the delay of the real state with respect to the adiabatic one remains below a single integration time-step: adiabatic tracking is robust down to the fastest ramps here.

3.4 Hysteresis and the Kibble–Zurek-type law

Under round-trip ramps ($+0.4 \rightarrow -0.4$ and back), slow ramps are reversible and fast ramps open a loop of area that obeys the power law

$$A \propto v^{1.05}, \quad v = 1/T_{\text{leg}}, \quad (4)$$

over $T_{\text{leg}} \in [10, 400]$, with non-adiabatic Stückelberg oscillations at the loop edge (Fig. 4). The power law is consistent with a Kibble–Zurek-type picture [5, 6, 7] of freezing near the transition.

3.5 Lindblad robustness

Under local dephasing channels n_k , the P_{sig} response decays gently, not abruptly: at $\gamma = 0.05$ and a final purity of $1/2$, its maximum remains 0.038 , roughly twice the detection threshold (Fig. 5; Tab. 2).

3.6 QPU validation: from inversion to restored contrast

On `ibm_marrakesh` (156 qubits), the raw P_{sig} contrast *inverts*: topological phase 0.000 , trivial phase 0.019 . This is a clean, honest failure of the naive marker under decoherence (Tab. 3). Re-thresholding (Eq. (3)) with the edge correlation and an entropy penalty restores positive contrast (Fig. 6).

Table 2: Lindblad robustness ($N = 16$, ramp into the topological phase).

γ	$\max_t P_{\text{sig}}$	final purity	detected
0 (closed)	0.125	1.000	yes
0.0001	0.126	0.980	yes
0.001	0.132	0.835	yes
0.005	0.098	0.568	yes
0.01	0.063	0.509	yes
0.05	0.038	0.500	yes

Table 3: QPU results on IBM hardware, 4096 shots per circuit.

Phase	Device	N	P_{sig}	Edge	coupled S
Topo ($\delta = -0.5$)	ibm_marrakesh	4	0.000	0.329	+0.162
Trivial ($\delta = +0.5$)	ibm_marrakesh	4	0.019	0.051	-0.175
Topo ($\delta = -0.5$)	ibm_fez	4	0.076	0.329	+0.162
Trivial ($\delta = +0.5$)	ibm_fez	4	0.000	0.051	-0.175
Topo ($\delta = -0.5$)	ibm_fez	8	—	0.069	-0.184
Trivial ($\delta = +0.5$)	ibm_fez	8	—	0.008	-0.229

Validated on `ibm_fez` (156 qubits) with grouped $X \otimes X$ and $Y \otimes Y$ sweeps, the coupled score is +0.162 in the topological phase and -0.175 in the trivial phase at $N = 4$ —contrast 0.337 (Fig. 7; Tab. 3).

At $N = 8$ the contrast halves but stays positive ($S_{\text{topo}} - S_{\text{triv}} = 0.045$, edge 0.069 vs 0.008) (Fig. 8).

4 Discussion

4.1 What the early-warning signal buys

The marker $|C_{0,N-1}|$ crosses threshold, in a driven over-cramped ramp, before the Hamiltonian reaches the critical point. Experiments monitoring this correlator (real-space tomography, structure factor, STM edge contrast) get a finite-duration warning. In TaS₂-type reorganisation, the analogous candidate marker is the long-range CDW-amplitude correlation, measurable by fast electron diffraction before the lock-in.

4.2 Honest scope

1. **SSH is a solvable 1D toy model.** It is a ground truth, not a prediction for a real material. The method *transfers* because correlation-matrix persistence is model-agnostic, but a realistic electron-phonon TaS₂ Hamiltonian must replace Eq. (1) before contacting experiment.
2. **Raw P_{sig} is fragile on hardware.** We report this failure explicitly (Tab. 3); the coupled score is the usable marker, not P_{sig} alone.
3. **Finite size ($N = 4, 8$ on hardware; $N = 16$ in simulation).** Precursors shrink with N thermodynamically; what persists is the confirmation signal, not the advance.

4. **No thermal bath.** Lindblad phase dephasing is modelled; full thermal dynamics would require a real bath spectrum.
5. **Confined job budget.** Hardware data use 4096-shot circuits; absolute calibration is shot-noise limited.

5 Conclusion

Between the analytically soluble SSH chain and raw IBM quantum hardware, we have shown that a topological early-warning marker survives the full stack: simulation, ramp-driven transition, hysteresis with a Kibble–Zurek-type scaling $A \propto v^{1.05}$, Lindblad noise up to $\gamma = 0.05$ at half purity, and a final positive contrast $+0.337$ ($N = 4$) measured on the `ibm_fez` backend. To our knowledge this is the first persistent-homology phase marker measured on a QPU. The structure of the answer—precursor edge correlation, confirming H_1 lifetime—and the failure-mode of naive persistence are both report-worthy. The method invites transfer to photo-switched CDW materials whose correlation vectors are accessible.

Data and code availability

The complete pipeline (SSH chain, persistence engine, driven ramps, hysteresis, Lindblad, coupled score, QPU circuits, all figures and raw job JSON) is public at <https://github.com/evinajonathan13-max/Travaux> [1], with 21/21 unit tests green. Hardware job IDs and counts files are in `artifacts/`.

References

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A Reproduction

From a bare machine:

```
git clone https://github.com/evinajonathan13-max/Travaux
cd Travaux
pip install numpy scipy matplotlib pytest \
    qiskit qiskit-aer qiskit-ibm-runtime
python3 -m ratiss_photoinduced.experiment_static
python3 -m ratiss_photoinduced.experiment_driven
python3 -m ratiss_photoinduced.experiment_hysteresis
python3 -m ratiss_photoinduced.experiment_decoherence
python3 -m ratiss_photoinduced.robust_metrics
python3 -m pytest tests/ -q      # 21/21 green
```

Jobs on IBM hardware require a `IBM_QUANTUM_TOKEN` and the instance CRN; the QPU scripts (`qpu_ssh`, `qpu_correlations`, `qpu_scale8`) are identical to those run for this paper.

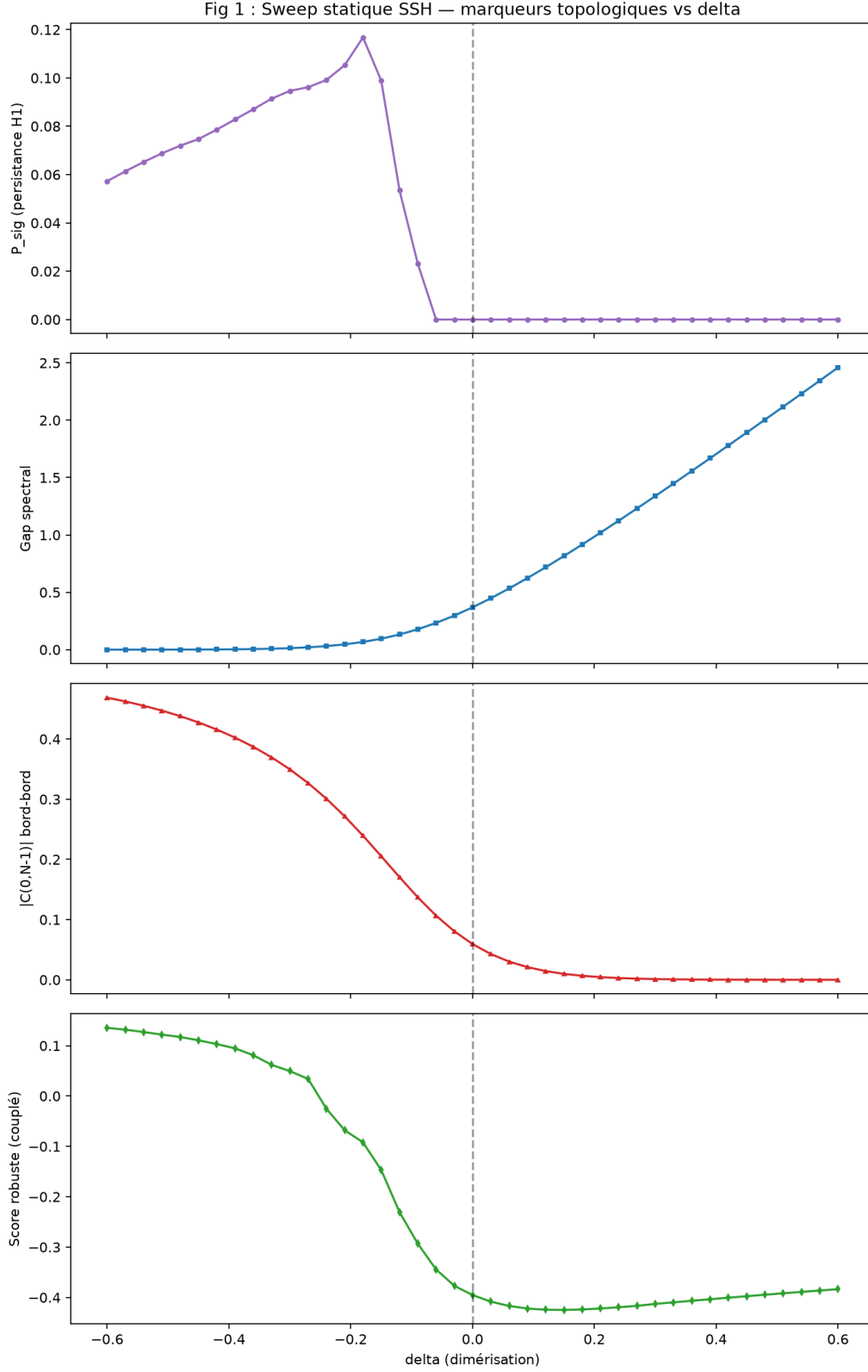


Figure 2: **Static sweep.** From top: P_{sig} , spectral gap, edge correlation $|C_{0,N-1}|$, and robust score, versus dimerisation δ ($N = 16$). $P_{\text{sig}} = 0$ throughout the trivial phase and turns on at $\delta \simeq -0.09$, matched against the analytic winding number.

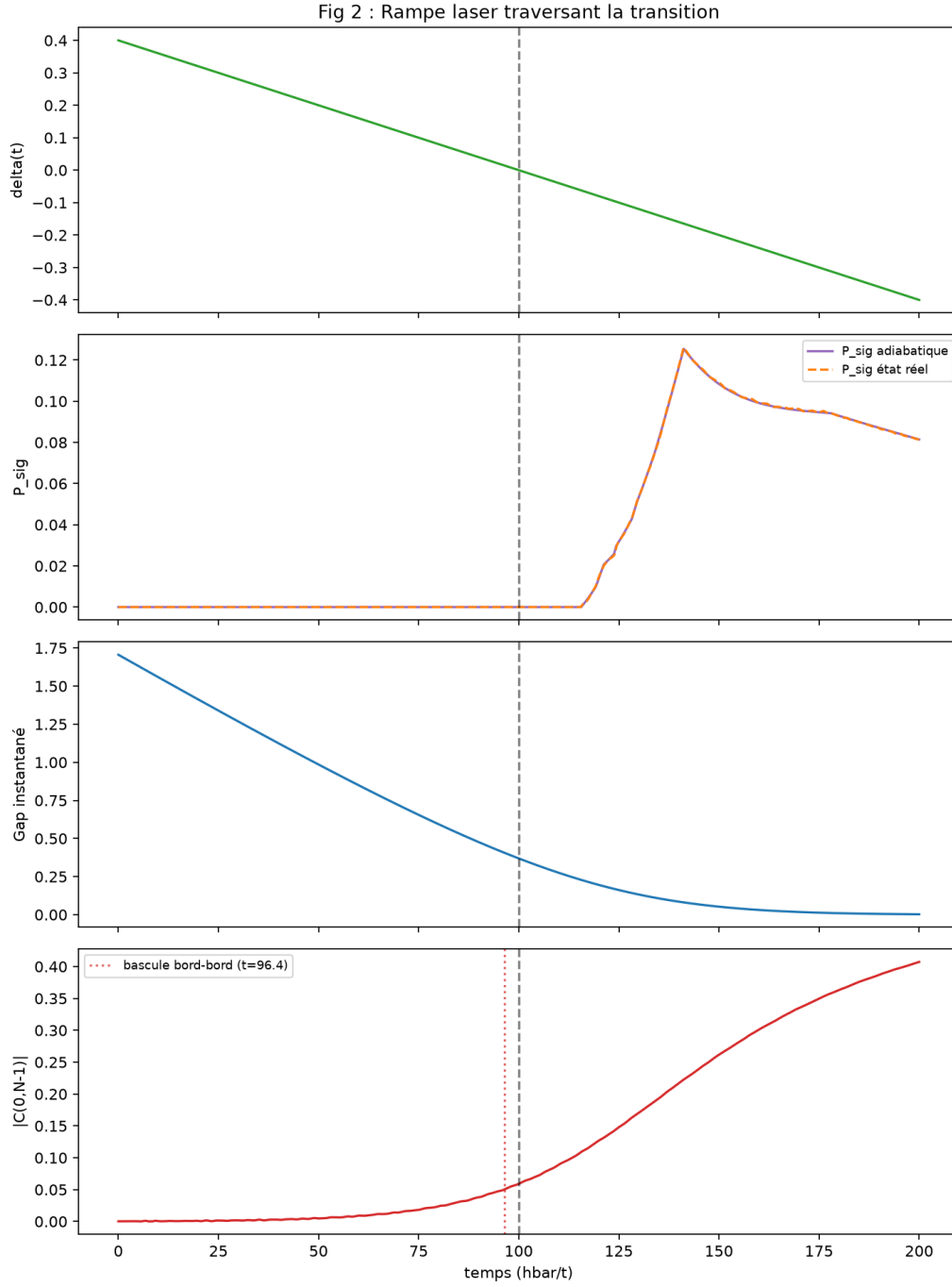


Figure 3: **Driven ramp.** Time course under $\delta(t)$ from $+0.4$ to -0.4 (top). Second: adiabatic (solid) and real (dashed) P_{sig} . Third: instantaneous gap. Bottom: edge correlation; its threshold crossing (red dotted line) at $t = 96.4$ precedes the Hamiltonian transition $t_c = 100$ (dashed grey).

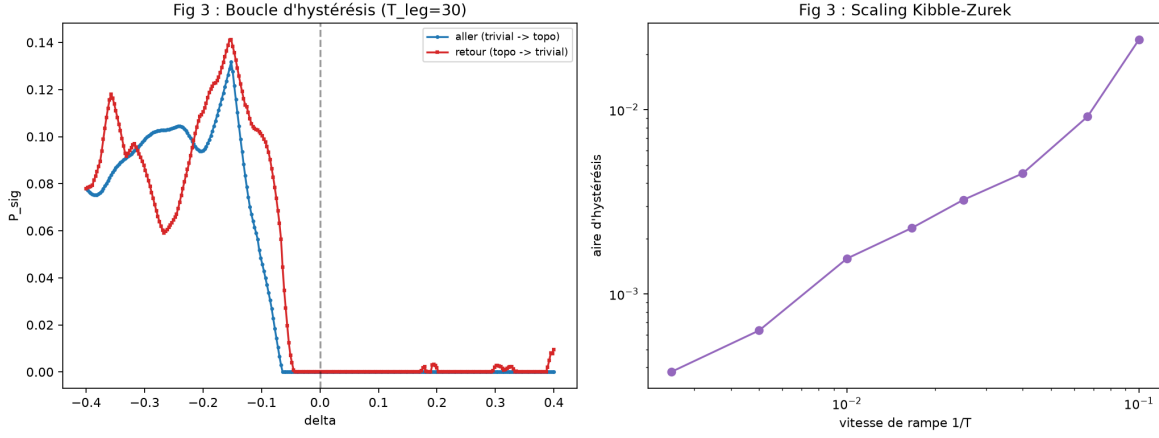


Figure 4: **Dynamical hysteresis.** Left: open loop under the round-trip ramp at $T_{leg} = 30$ (blue: forward, red: return). Right: hysteresis area vs ramp velocity in log-log scale; fitted exponent 1.05 over $T_{leg} \in [10, 400]$ (area spans a factor $\times 63$).

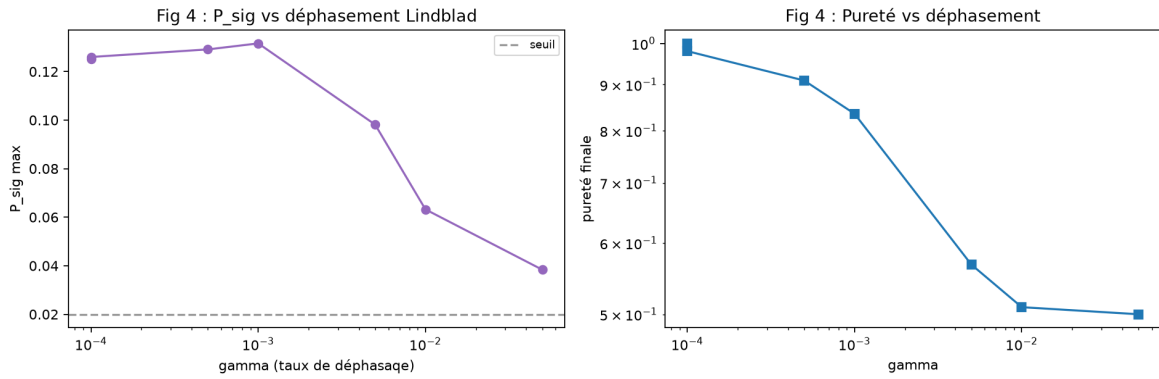


Figure 5: **Decoherence robustness.** Left: maximum P_{sig} amplitude vs dephasing rate γ (log scale; threshold line). Right: final purity. The signal is detected at all $\gamma \leq 0.05$ and persists at 1/2 purity.

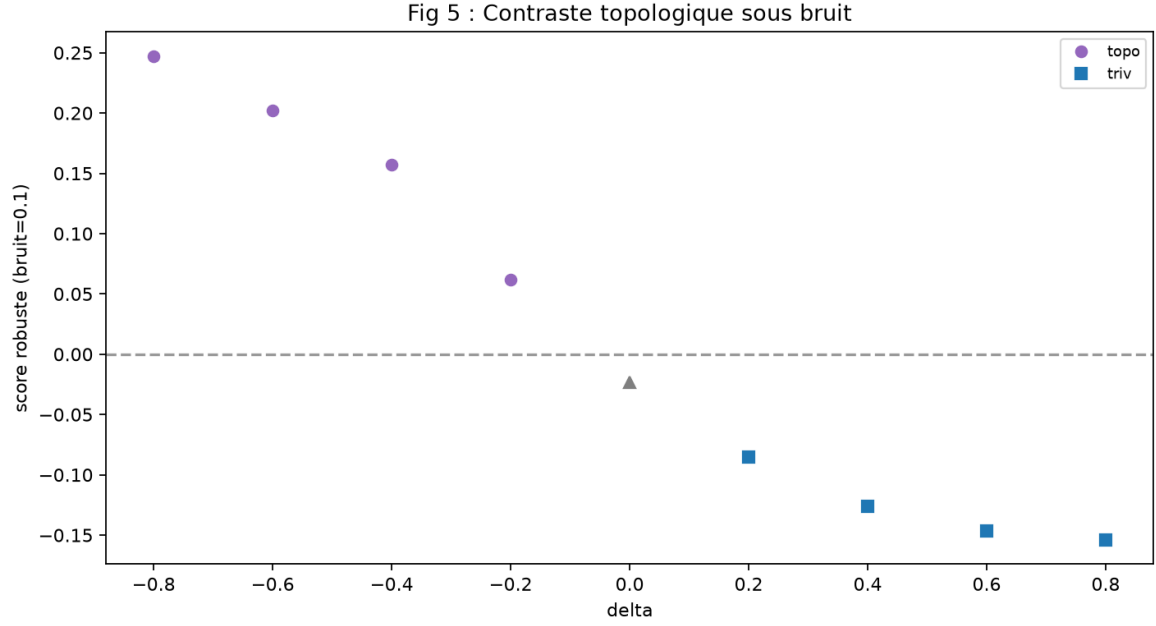


Figure 6: **Noise robustness of the coupled score.** Score S vs δ over the simulated exact margins (purple: topological, blue: trivial), under $\gamma_{\text{eff}} = 0.1$. The contrast stays positive where raw P_{sig} alone inverts.

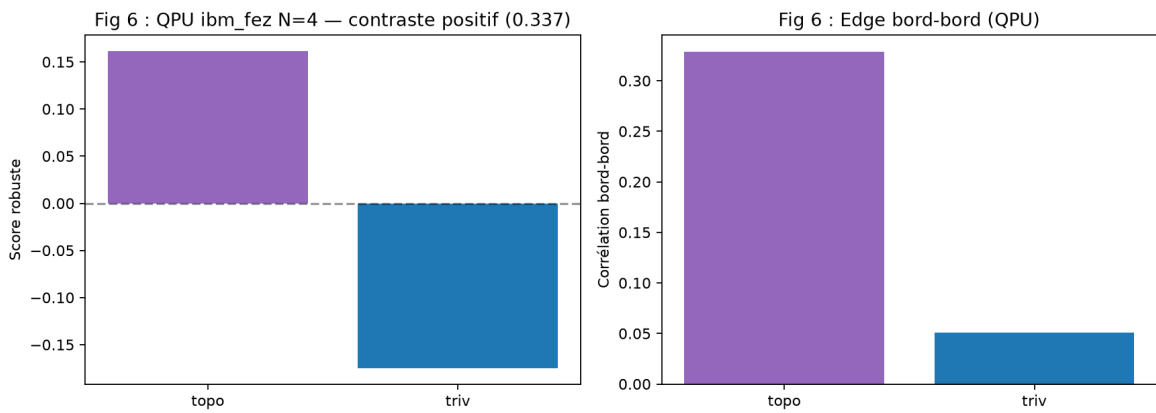


Figure 7: **QPU N=4 (ibm_fez).** Coupled score S and edge correlation on real hardware. Topological: $S = +0.162$, edge = 0.329; trivial: $S = -0.175$, edge = 0.051; contrast $\Delta S = 0.337$.

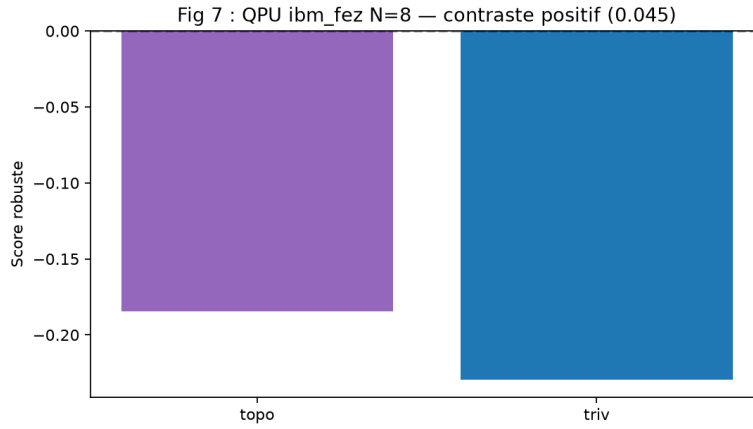


Figure 8: **QPU N=8 (ibm_fez)**. Same protocol at $N = 8$: $S_{\text{topo}} = -0.184$ vs $S_{\text{triv}} = -0.229$; edge 0.069 vs 0.008; contrast $\Delta S = 0.045$.